

# CreditWise Loan Approval Prediction System

## Project Report

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### 1. Project Overview

CreditWise is a machine learning-based loan approval prediction system designed to assist financial institutions in making faster and more consistent decisions. The project uses historical applicant data to predict whether a loan should be approved or rejected.

### 2. Problem Statement

Manual loan verification is time-consuming and prone to inconsistencies. The goal of this system is to reduce human effort while improving decision consistency and reducing financial risk.

### 3. Dataset

The project uses a structured dataset containing applicant demographics, financial information, loan details, and the target variable `Loan_Approved`. The dataset includes both numerical and categorical attributes.

### 4. Methodology

- Loaded and inspected the loan dataset using pandas.
- Handled missing values through imputation for both numerical and categorical columns.
- Performed exploratory data analysis using visualizations such as pie charts, histograms, and boxplots.
- Engineered new features such as squared DTI and Credit Score terms.
- Encoded categorical variables and scaled features for modeling.
- Trained multiple classifiers and compared their performance.

### 5. Models and Results

The project compared Logistic Regression, K-Nearest Neighbors, and Gaussian Naive Bayes. Logistic Regression produced the best overall performance with strong precision and recall balance.

Observed metrics: Accuracy 0.916, Precision 0.881, Recall 0.852, and F1-score 0.867.

### 6. Streamlit Deployment

The project was also deployed as a Streamlit web application so users can input applicant details and receive an instant loan approval prediction through the trained model.

### 7. Key Insights

Credit Score and Debt-to-Income Ratio were identified as more influential predictors than income alone. This highlights the importance of repayment behavior and financial discipline in loan approval decisions.

## 8. Conclusion

CreditWise demonstrates an end-to-end machine learning workflow for loan risk assessment. It combines data preprocessing, visualization, model training, evaluation, and deployment in a practical and user-friendly format.

## Appendix

Files included in the project: `credit_wise.ipynb`, `loan_approval_data.csv`, `app.py`, `requirements.txt`, and this report.